

Mismatch-Click Defects and Replicated Update Maps in an Adaptive Class-A Fermion Circuit: A Working Technical Note

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We investigate whether the binary signs in the perfect-correction covariance map admit a useful dynamical-defect or statistical-mechanics description. The target sign is fixed by the controller band, while the Born sign defines a binary mismatch click $d = (1 - \eta_m \eta_s)/2$. We show that this click is an exact centered Bernoulli innovation driving a rank-at-most-one covariance kick. The measured orbital is healed after either outcome; for a pure Slater trajectory, the branch label instead flips one state-dependent partner orbital. This yields an exact positive Perron–Frobenius transfer operator on the full covariance manifold, a Gaussian-closed tilted channel for the full counting statistics of mismatch activity, and a local replicated density-operator transfer whose trace is $\sum_{\xi} p_{\xi}^n$. Two marked replicas represent the Born-averaged squared correlator and intrinsic charge variance. The one-copy covariance average closes affinely, whereas its exact fluctuation equation contains a nonlinear shot-noise source and generates an infinite moment hierarchy. In contrast to the classical cellular automaton of O’Dea *et al.*, overlapping noncommuting controller projectors do not preserve a common diagonal algebra; the click record alone is therefore not generically an autonomous local Markov process. We finish with a testable narrow-distribution closure and a retarded effective click action whose kernel is a linearized covariance Green function. This is a working note: exact identities, proposed approximations, and conjectural wall physics are labeled separately.

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		The natural binary field is their relative sign,	
		$d = \frac{1 - \eta_m \eta_s}{2} \in \{0, 1\}. \quad (2)$	
		It records whether the measurement disagrees with the	
		target and therefore whether the fresh-ancilla correction	

fires. The provisional answer developed in this note has four levels.

1. *Exact classical hidden-state process.* The covariance G is a classical continuous state, and d is a binary emission from a positive Markov kernel on the covariance manifold.
2. *Exact one-copy closure.* After Born averaging, G obeys a deterministic affine reset map. The mean mismatch rate is consequently one-copy closed.
3. *Exact replica transfer.* Shared outcomes couple density-operator replicas. The resulting local transfer gives record Rényi sums and marked formulas for squared correlators and quantum density fluctuations.
4. *No generic click-only cellular automaton.* The click is healed at the measured orbital, and its effect propagates through a state-dependent partner mode. Eliminating G produces retarded many-body interactions among clicks rather than a finite-range autonomous bit rule.

The fourth point is not a failure of the statistical-mechanics idea. It says that the exact theory is either a positive infinite-dimensional classical transfer problem or a local interacting replicated-fermion tensor network. A local scalar defect theory, if present at the wall, must emerge after a controlled coarse graining.

II. MICROSCOPIC INSTRUMENT AND SIGN DICTIONARY

Consider a parity-even, number-conserving Gaussian state of N complex fermion modes. We use

$$C_{ij} = \text{Tr}(\rho c_i^\dagger c_j), \quad G = 2C - \mathbf{1}, \quad -\mathbf{1} \leq G \leq \mathbf{1}. \quad (3)$$

For a normalized controller orbital $a_\chi = \sum_i \chi_i c_i$, define

$$n_\chi = a_\chi^\dagger a_\chi, \quad P = \chi \chi^\dagger, \quad Q = \mathbf{1} - P, \quad g = \chi^\dagger G \chi. \quad (4)$$

The measurement projectors are $\Pi_1 = n_\chi$ and $\Pi_0 = \mathbf{1} - n_\chi$. For a target $s \in \{0, 1\}$, write

$$\tau \equiv \eta_s = 2s - 1. \quad (5)$$

The perfect-correction Kraus operators can then be organized by the mismatch click rather than by the raw outcome:

$$K_{0|s} = \Pi_s, \quad K_{1|s} = (a_\chi^\dagger)^s a_\chi^{1-s}, \quad (6)$$

$$K_{d|s}^\dagger K_{d|s} = \Pi_{s \oplus d}.$$

The raw measurement sign is recovered from

$$\eta_m = \tau(-1)^d. \quad (7)$$

The two branch probabilities are

$$p_0(G) = \frac{1 + \tau g}{2}, \quad p_1(G) = \frac{1 - \tau g}{2}. \quad (8)$$

We will repeatedly use the *soft mismatch propensity*

$$q_s(G) \equiv p_1(G) = \frac{1 - \tau \chi^\dagger G \chi}{2}. \quad (9)$$

The hard click is a Bernoulli sample $d \sim \text{Bernoulli}(q_s(G))$.

The correction has a signed charge. With $\hat{N} = \sum_i c_i^\dagger c_i$,

$$[\hat{N}, K_{d|s}] = \tau d K_{d|s}. \quad (10)$$

Consequently, for an input branch of definite particle number,

$$N_T - N_0 = \sum_{t=1}^D \tau_t d_t. \quad (11)$$

A lower-band mismatch is a particle injection and an upper-band mismatch is a hole injection. For number-indefinite mixed inputs, Eq. (10) remains the operator statement, but conditioning can also change the mean charge of the unmeasured sector; Eq. (11) should then not be applied as a branchwise expectation-value identity.

III. THE MISMATCH CLICK AS AN EXACT COVARIANCE INNOVATION

Define two matrices built from the incoming covariance,

$$B(G) = QGQ + \tau P, \quad R(G) = QGPGQ = (QG\chi)(QG\chi)^\dagger. \quad (12)$$

For $|g| < 1$, the rank-one resolvent in Eq. (1) gives

$$\Phi_0(G) = B(G) - \frac{\tau R(G)}{1 + \tau g}, \quad \Phi_1(G) = B(G) + \frac{\tau R(G)}{1 - \tau g}. \quad (13)$$

Equivalently,

$$\Phi_d(G) = B(G) + \frac{2\tau [d - q_s(G)]}{1 - g^2} R(G). \quad (14)$$

Thus

$$\xi_d \equiv d - q_s(G), \quad \mathbb{E}[\xi_d|G] = 0, \quad \mathbb{E}[\xi_d^2|G] = \frac{1 - g^2}{4} \quad (15)$$

is an exact martingale innovation. For every physical covariance,

$$\text{Tr } R = \chi^\dagger G Q G \chi \leq 1 - g^2. \quad (16)$$

At $|g| = 1$, one branch has zero probability, $R = 0$, and the unique realized update is $\Phi = B$. The normalized

TABLE I. Claim ledger and roadmap for the mismatch-defect construction.

Status	Statement	Required qualification or test
Exact	$d = (1 - \eta_m \eta_s)/2$ is a Bernoulli innovation and drives the rank-at-most-one covariance kick in Eq. (14).	Only nonzero-probability normalized branches are defined; the realized endpoint update and Born-weighted moment identities are regular.
Exact	The physical covariance mean closes affinely, and the covariance fluctuation obeys Eq. (46).	The second equation is not a finite moment closure because its source depends on the full covariance law.
Exact	Click activity has the Gaussian tilted recursion Eq. (53); record Rényi sums have the shared replica transfer Eq. (57).	These are different tilted objects and must not be interchanged.
Formal	Two marked replicas followed by $n \rightarrow 1$ produce physical squared conditional observables.	Continuing an integer-replica transfer, especially after a large-size limit, must be fixed by a transfer theory or checked against an independent estimator.
Conditional	A common-basis classical automaton exists if $\Delta\mathcal{E} = \Delta\mathcal{E}\Delta$, preferably instrument by instrument.	Generic overlapping projectors violate the natural local diagonal closure.
Approximate	Freezing the nonlinear noise source gives a discrete Lyapunov equation; a separate weak-noise linearization gives the retarded action in Eq. (78).	Validate both against direct source and conditional-rate measurements; retain higher-body residuals.
Conjectural	Slow wall modes mediate a long-ranged, possibly directional defect interaction.	Requires wall/bulk Green-function comparison, size scaling, and successful prediction of click correlations.

counterfactual branch is not defined. Born-weighted expressions containing $R^{\otimes 2}/(1 - g^2)$ have the unique zero limit by Eq. (16), whereas $R/(1 - g^2)$ and its orientation need not have a path-independent limit.

The click is not a persistent defect of the just-measured mode. Every nonzero-probability branch satisfies

$$\Phi_d(G)\chi = \tau\chi, \quad \chi^\dagger \Phi_d(G)\chi = \tau. \quad (17)$$

The mode χ factorizes with its target occupation immediately after the event. Away from the deterministic endpoints, the record survives in the orthogonal complement through the rank-at-most-one matrix R .

A. Pure-state partner orbital

For a pure Slater trajectory, $G^2 = \mathbf{1}$, Eq. (16) is saturated. Whenever $|g| < 1$, define

$$u = \frac{QG\chi}{\sqrt{1 - g^2}}, \quad U = uu^\dagger. \quad (18)$$

Then u is normalized, orthogonal to χ , and

$$R = (1 - g^2)U, \quad QGQu = -gu. \quad (19)$$

Equation (14) becomes

$$\Phi_d(G) = B(G) + 2\tau[d - q_s(G)]U. \quad (20)$$

On the two-dimensional active subspace,

$$\Phi_d(G)\chi = \tau\chi, \quad \Phi_d(G)u = \tau(2d - 1)u. \quad (21)$$

All one-particle directions outside this active plane are identical in the two branches. Therefore

$$\Phi_1(G) - \Phi_0(G) = 2\tau U. \quad (22)$$

Microscopically, the correction heals χ , while $d = 1$ flips exactly one state-dependent Schmidt-partner orbital relative to the matched branch. This is the most literal meaning of a dynamical particle/hole defect available at the Gaussian level. A worldline description would have to track the evolving orientation u_t , not only the bits d_t .

IV. HOW A CLICK INFLUENCES THE NEXT CONTROLLER

Assume $|g| < 1$, and let the next measured orbital be ψ , with target sign τ' . Its mismatch propensity after the present event is

$$q'_d = \frac{1 - \tau'\psi^\dagger \Phi_d(G)\psi}{2}. \quad (23)$$

Set

$$q'_B = \frac{1 - \tau'\psi^\dagger B(G)\psi}{2}, \quad w_{\psi \leftarrow \chi}(G) = \frac{\psi^\dagger R(G)\psi}{1 - g^2} = \frac{|\psi^\dagger QG\chi|^2}{1 - g^2}. \quad (24)$$

The bound in Eq. (16) implies $0 \leq w \leq 1$. Direct substitution gives the exact propagation law

$$\boxed{q'_d = q'_B - \tau\tau'[d - q_s(G)]w_{\psi \leftarrow \chi}(G)}. \quad (25)$$

In particular,

$$q'_1 - q'_0 = -\tau\tau'w_{\psi \leftarrow \chi}(G). \quad (26)$$

A mismatch suppresses the next mismatch for equal target signs and enhances it for opposite target signs. For a pure incoming state, $w = |\langle \psi | u \rangle|^2$. Equation (25) looks like a two-state branching rule, but its rate depends on the full incoming covariance through both u and g . After marginalizing over G , the click string is therefore generally non-Markovian.

A. A geometric frustration floor

The nonorthogonality of opposite-target orbitals provides an outcome-independent source of future clicks.

Proposition 1 (Consecutive opposite-target bound). *Suppose an update has just reset χ to sign τ , so that $G\chi = \tau\chi$. Let the immediately following controller orbital ψ have target $\tau' = -\tau$. Then*

$$q_{\psi, \tau'}(G) \geq |\langle \chi | \psi \rangle|^2. \quad (27)$$

The proof is given in Appendix C. This bound suggests a direct controller diagnostic: construct the directed conflict graph of the actual chronological word, weight an edge $j \rightarrow k$ by $|\langle \chi_j | \chi_k \rangle|^2$ when $\tau_j \tau_k = -1$, and compare its large weights with the observed mismatch-click profile. The theorem supplies a lower bound for consecutive updates; it does not assert that every nonconsecutive pair contributes additively.

A covariance with zero mismatch propensity for every controller would have to satisfy

$$G_* \chi_j = \tau_j \chi_j \quad \text{for every } j. \quad (28)$$

Since eigenvectors of a Hermitian matrix with eigenvalues $+1$ and -1 are orthogonal, Eq. (28) is impossible whenever two nonorthogonal controller modes carry opposite targets. Thus cross-target overlap is a precise obstruction to a common zero-click state. Whether this obstruction is present, and where it is localized, must be evaluated for the actual finite-support controller; it is not inferred here from topology alone.

V. EXACT POSITIVE TRANSFER ON THE COVARIANCE MANIFOLD

Fix the complete chronological schedule $(\chi_t, \tau_t)_{t=1}^D$. The covariance is a classical hidden state and the mismatch click is its binary emission:

$$d_t \sim p_{t,d}(G_{t-1}), \quad G_t = \Phi_{t,d_t}(G_{t-1}). \quad (29)$$

The corresponding Markov kernel is

$$\mathcal{L}_{t,1}(G', G) = \sum_{d=0,1} p_{t,d}(G) \delta(G' - \Phi_{t,d}(G)). \quad (30)$$

It is normalized in G' and defines an exact Perron–Frobenius pushforward on probability measures over the admissible covariance manifold $\mathcal{X}$. The dual Koopman operator acts on an observable $f(G)$ as

$$(\mathcal{K}_t f)(G) = \sum_{d=0,1} p_{t,d}(G) f(\Phi_{t,d}(G)). \quad (31)$$

The complete click-string probability is

$$p[\mathbf{d}] = \prod_{t=1}^D p_{t,d_t}(G_{t-1}[d_1, \dots, d_{t-1}]). \quad (32)$$

This already defines a positive classical statistical weight. A particularly transparent Ising form follows from

$$\sigma_t = (-1)^{d_t} = \eta_{m_t} \tau_t, \quad h_t = \operatorname{arctanh}(\tau_t g_t). \quad (33)$$

Then

$$p(d_t | G_{t-1}) = \frac{e^{h_t \sigma_t}}{2 \cosh h_t}, \quad (34)$$

and

$$-\log p[\mathbf{d}] = \sum_t \{ \log[2 \cosh h_t[\mathbf{d}_{<t}]] - h_t[\mathbf{d}_{<t}] \sigma_t \}. \quad (35)$$

It is literally an Ising history in a causal self-consistent field. The price of eliminating G is that h_t is a nonlinear functional of all earlier spins. The band signs τ_t form a fixed, generally staggered spacetime background.

For real $n > 0$, define the positive replica-tilted covariance kernel

$$\mathcal{L}_{t,n}(G', G) = \sum_{d=0,1} p_{t,d}(G)^n \delta(G' - \Phi_{t,d}(G)). \quad (36)$$

Its total mass after the full word is

$$\boxed{\mathcal{Z}_n = \langle 1 | \mathcal{L}_{D,n} \cdots \mathcal{L}_{1,n} | G_0 \rangle = \sum_{\mathbf{d}} p[\mathbf{d}]^n}. \quad (37)$$

At $n = 1$, Eq. (36) is stochastic. At $n \neq 1$, it is positive but not mass preserving. Normalizing $p_d(G)^n$ independently at each time does *not* produce the global replica-biased process, because the future weight depends on the present covariance. A normalized dynamics requires the usual backward harmonic function or Doob transform of the complete tilted transfer.

For a causal record the Shannon entropy also obeys the exact chain rule

$$H_{\text{rec}} = -\partial_n \log \mathcal{Z}_n|_{n=1} = \sum_t \mathbb{E}[h_2(q_t)], \quad (38)$$

where $h_2(q) = -q \log q - (1-q) \log(1-q)$. Since h_2 is nonlinear, H_{rec} requires the distribution of the propensities, not merely their mean.

VI. WHAT CLOSES AFTER PHYSICAL BORN AVERAGING

The innovation form makes the exceptional simplicity of $n = 1$ explicit:

$$\mathbb{E}_d[\Phi_d(G)|G] = B(G) = QGQ + \tau P. \quad (39)$$

The averaged density-matrix channel is the fermionic replacement channel

$$\mathcal{E}_s(\rho) = \sum_d K_{d|s} \rho K_{d|s}^\dagger = |s\rangle\langle s|_\chi \hat{\otimes} \text{Tr}_\chi \rho, \quad (40)$$

and its covariance action is Eq. (39). Because B is affine,

$$\bar{G}_t = Q_t \bar{G}_{t-1} Q_t + \tau_t P_t \quad (41)$$

for a fixed schedule. In particular, the mean mismatch rate is one-copy closed:

$$\mathbb{E} d_t = \frac{1 - \tau_t \chi_t^\dagger \bar{G}_{t-1} \chi_t}{2}. \quad (42)$$

The mean signed correction rate is

$$\mathbb{E}(\tau_t d_t) = \frac{\tau_t - \chi_t^\dagger \bar{G}_{t-1} \chi_t}{2} = \frac{\text{Tr} \bar{G}_t - \text{Tr} \bar{G}_{t-1}}{2}. \quad (43)$$

Hence its sum over a stationary complete cycle vanishes, even though the unsigned activity can remain finite.

A. Exact covariance-fluctuation equation

The conditional second moment is

$$\mathbb{E}_d[\Phi_d(G)^{\otimes 2}|G] = B(G)^{\otimes 2} + \frac{R(G)^{\otimes 2}}{1 - g^2}. \quad (44)$$

Thus a mismatch click is a rank-at-most-one shot-noise source. To write a positive matrix covariance, use column-stacking vectorization of one-body operators and define

$$\Sigma_t = \mathbb{E}[|\delta G_t\rangle\langle\delta G_t|], \quad \delta G_t = G_t - \bar{G}_t. \quad (45)$$

Let $\mathcal{A}_t(H) = Q_t H Q_t$ and $\hat{\mathcal{A}}_t = Q_t^* \otimes Q_t$. Since the innovation has zero conditional mean, all cross terms with the past vanish and one obtains

$$\Sigma_t = \hat{\mathcal{A}}_t \Sigma_{t-1} \hat{\mathcal{A}}_t^\dagger + \mathbb{E}\left[\frac{|R_t\rangle\langle R_t|}{1 - g_t^2}\right]. \quad (46)$$

Equation (46) is the exact averaged replica-like update sought for density fluctuations. It is not a finite closure: the source is quartic and rational in the random incoming covariance. On any interior domain $|g| \leq 1 - \epsilon$, expanding

$(1 - g^2)^{-1} = \sum_{k \geq 0} g^{2k}$ displays dependence on moments of arbitrarily high order. The expansion is not uniform at the endpoints, but the source itself has the regular zero limit because $\|R\|_{\text{HS}}^2/(1 - g^2) \leq 1 - g^2$. Special twirls or large-flavor limits may close this hierarchy, but generic Gaussian purity does not.

For a region A , the within-trajectory quantum charge variance is

$$F_{A,\xi}^{(q)} = \text{tr}[C_{\xi,A}(\mathbf{1}_A - C_{\xi,A})] = \frac{|A| - \text{tr} G_{\xi,A}^2}{4}. \quad (47)$$

It is therefore fixed by a contraction of the second covariance moment, not by $\bar{G}$. Likewise, for $i \neq j$,

$$\overline{\langle \delta n_i \delta n_j \rangle_\xi} = -\overline{|C_{\xi,ij}|^2}. \quad (48)$$

The distinction between $\overline{|C_{ij}|^2}$ and $|\overline{C_{ij}}|^2$ is precisely the trajectory-fluctuation contribution. Charge cumulants and free-fermion fluctuation-entanglement relations provide the broader context for these observables [2, 3].

VII. A GAUSSIAN-CLOSED TILTED CHANNEL FOR CLICK ACTIVITY

The full counting statistics of the mismatch clicks has a simpler exact construction than same-record nonlinear state observables. Give a click a fugacity x and define

$$\mathcal{E}_{s,x}(\rho) = \sum_{d=0,1} x^d K_{d|s} \rho K_{d|s}^\dagger. \quad (49)$$

In the Fock factorization associated with χ ,

$$\begin{aligned} \mathcal{E}_{s,x}(\rho) &= |s\rangle\langle s|_\chi \hat{\otimes} \text{Tr}_\chi[W_s(x)\rho], \\ W_s(x) &= \Pi_s + x\Pi_{1-s} = x^{\Pi_{1-s}}. \end{aligned} \quad (50)$$

For $x > 0$, $W_s(x)$ is the exponential of a quadratic number operator. Products and partial traces of fermionic Gaussian operators remain Gaussian; therefore the normalized version of Eq. (49) closes exactly on a covariance, despite being non-trace-preserving before normalization.

For a normalized incoming Gaussian state, the one-step normalization is

$$z_x(G) = p_0(G) + x p_1(G) = \frac{1 + \tau g + x(1 - \tau g)}{2}. \quad (51)$$

The normalized tilted covariance is

$$\Psi_{s,x}(G) = B(G) + \tau \frac{x - 1}{1 + \tau g + x(1 - \tau g)} R(G). \quad (52)$$

The limits provide useful checks: $x = 1$ gives the unconditional mean map, $x = 0$ gives the matched branch, and $x \rightarrow \infty$ gives the mismatch branch, provided the selected branch has nonzero Born weight. A normalized zero-probability branch is undefined.

For event-dependent sources x_t , initialize $Z_0 = 1$ and the physical G_0 , then iterate

$$Z_t = Z_{t-1} z_{x_t}(G_{t-1}^{\mathbf{x}}), \quad G_t^{\mathbf{x}} = \Psi_{s_t, x_t}(G_{t-1}^{\mathbf{x}}). \quad (53)$$

This gives exactly

$$Z_D(\mathbf{x}) = \sum_{\mathbf{d}} p[\mathbf{d}] \prod_t x_t^{d_t}. \quad (54)$$

Taking $x_t = e^{\lambda_t}$ generates all multitime click cumulants. Taking $x_t = e^{\lambda + i\theta\tau_t}$ jointly counts unsigned activity and signed correction charge. This is the natural dynamical-activity ensemble of quantum-trajectory large-deviation theory [4, 5]. For complex sources the tilted Gaussian operator and its covariance are generally complex; the normalized recursion is an analytic continuation valid away from zeros of the intermediate normalization, while the unnormalized generating polynomial remains the primary object. For a periodically repeated autonomous cycle, the leading eigenvalue of the unnormalized cycle superoperator gives the scaled cumulant-generating function. A randomly changing schedule requires a separate quenched or annealed convention.

VIII. REPLICATED DENSITY TRANSFER AND MARKED OBSERVABLES

The activity tilt weights a physical record by p_ξ . Record Rényi sums and squared conditional observables require shared-record replicas. Define the unnormalized branch state

$$\tilde{\rho}_\xi = K_\xi \rho_0 K_\xi^\dagger, \quad p_\xi = \text{Tr } \tilde{\rho}_\xi, \quad \rho_\xi = \frac{\tilde{\rho}_\xi}{p_\xi}. \quad (55)$$

For integer n , introduce

$$\Omega_n(\mathbf{x}) = \sum_{\xi} \left(\prod_t x_t^{d_t(\xi)} \right) \tilde{\rho}_\xi^{\otimes n}. \quad (56)$$

Here replica Hilbert spaces are distinguishable and use the ordinary tensor product; $\hat{\otimes}$ is reserved for fermionic mode factorizations within one replica. Let $\mathcal{K}_{t,d}(X) = K_{t,d} X K_{t,d}^\dagger$. The exact local replicated update is

$$\boxed{\mathcal{T}_t^{(n)}(x_t) = \mathcal{K}_{t,0}^{\otimes n} + x_t \mathcal{K}_{t,1}^{\otimes n}, \quad \Omega_{n,t} = \mathcal{T}_t^{(n)}(x_t) \Omega_{n,t-1}.} \quad (57)$$

The same click d_t is shared by every replica. Summing the outcome independently in each replica would instead construct powers of the unconditional state and would erase the same-record correlations. The trace is

$$\boxed{Z_n(\mathbf{x}) = \text{Tr } \Omega_n(\mathbf{x}) = \sum_{\xi} p_\xi^n \prod_t x_t^{d_t(\xi)}.} \quad (58)$$

At $\mathbf{x} = 1$, this is the record partition function used in transfer-matrix studies of monitored criticality [6–8]. In vectorized form, each event is a local tensor

$$T_t^{(n)}(x) = \sum_{d=0,1} x^d (K_{t,d}^* \otimes K_{t,d})^{\otimes n}. \quad (59)$$

Thus d_t is a spacetime Ising variable coupled to replicated fermions. Integrating out the fermions gives the positive weight $p[\mathbf{d}]^n$, but generically produces a nonlocal many-body action for $\mathbf{d}$.

The special cancellation at $n = 1$ can be seen locally for one normalized incoming Gaussian state. Let $a = p_0 = (1 + \tau g)/2$ and $b = p_1 = (1 - \tau g)/2$. Then

$$\sum_d p_d^n \Phi_d = (a^n + b^n)B + \frac{\tau}{2}(b^{n-1} - a^{n-1})R, \quad (60)$$

$$\begin{aligned} \sum_d p_d^n \Phi_d^{\otimes 2} &= (a^n + b^n)B^{\otimes 2} \\ &\quad + \frac{\tau}{2}(b^{n-1} - a^{n-1})(B \otimes R + R \otimes B) \\ &\quad + \frac{1}{4}(a^{n-2} + b^{n-2})R^{\otimes 2}. \end{aligned} \quad (61)$$

Only at $n = 1$ does the first-moment R term vanish. These are local conditional sums, not an update rule for a general correlated $\Omega_{n,t-1}$ produced by earlier shared outcomes; future tilted weights still enter the normalized global n -ensemble. When a or b vanishes, the complete Born-weighted expressions—not their individual negative-power factors—are taken to the endpoint.

A. Two marked replicas

Set $\mathbf{x} = \mathbf{1}$ in this subsection and abbreviate $\Omega_n = \Omega_n(\mathbf{1})$. For integer $n \geq k$, insert observables on k distinguished replicas and leave the remaining replicas unmarked. One obtains

$$\frac{\text{Tr}[(O_1^{(1)} \cdots O_k^{(k)})\Omega_n]}{Z_n} = \frac{\sum_{\xi} p_\xi^n \prod_{a=1}^k \langle O_a \rangle_{\xi}}{\sum_{\xi} p_\xi^n}. \quad (62)$$

For a finite record space, the right-hand side defines a canonical real- n function near $n = 1$. The replica prescription recovers the physical Born ensemble by continuing the integer-replica transfer to $n \rightarrow 1$. In particular,

$$\boxed{|\overline{C_{\xi,ij}}|^2 = \lim_{n \rightarrow 1} \frac{\text{Tr}[(c_i^\dagger c_j)^{(1)}(c_j^\dagger c_i)^{(2)}\Omega_n]}{Z_n}.} \quad (63)$$

The within-trajectory quantum charge variance is

$$\boxed{F_{A,\xi}^{(q)} = \lim_{n \rightarrow 1} \frac{\text{Tr}[(N_A^{(2)})^{(1)} - N_A^{(1)} N_A^{(2)}]\Omega_n]}{Z_n}.} \quad (64)$$

Likewise,

$$\boxed{\overline{\langle \delta n_i \delta n_j \rangle}_{\xi} = \lim_{n \rightarrow 1} \frac{\text{Tr}[(n_i n_j)^{(1)} - n_i^{(1)} n_j^{(2)}]\Omega_n]}{Z_n}.} \quad (65)$$

What is formal is reconstructing that continuation from integer-replica data, especially after taking a large-depth or thermodynamic limit; a transfer spectrum, field theory, or independent estimator must fix it. Also note that Ω_2 carries p_ξ^2 , not p_ξ : simply simulating two replicas at $n = 2$ does not directly give the physical Born-averaged squared correlator.

IX. WHY THE O'DEA CLASSICAL AUTOMATON DOES NOT TRANSFER DIRECTLY

O'Dea, Morningstar, Gopalakrishnan, and Khemani obtain a classical stochastic process for diagonal elements of the trajectory-averaged density matrix in their adaptive spin circuits [9]. Their doubled Hilbert space is the bra/ket vectorization of an $n = 1$ channel, not the same-record replica in Eq. (56). A constrained Haar average preserves a common diagonal algebra and kills the off-diagonal block after a unitary layer. The surviving populations are therefore genuine classical bitstrings whose down spins are time-slice configuration defects carried between updates. In their cluster-state construction, a finite-depth Clifford transformation first maps commuting stabilizer violations to onsite bits.

There is a concise general criterion. Let Δ dephase in a proposed product basis and let $\mathcal{E}$ be a quantum channel. Its diagonal populations obey an autonomous classical update exactly when

$$\boxed{\Delta\mathcal{E} = \Delta\mathcal{E}\Delta}. \quad (66)$$

If $\{K_a\}$ are Kraus operators, Eq. (66) is equivalent to

$$\sum_a (K_a)_{xy} (K_a)_{xz}^* = 0 \quad \text{for every } x \text{ and } y \neq z. \quad (67)$$

Then

$$T_{xy} = \sum_a |(K_a)_{xy}|^2 \geq 0, \quad \sum_x T_{xy} = 1 \quad (68)$$

is a stochastic matrix. A record-resolved classical instrument needs the stronger condition $\Delta\mathcal{E}_d = \Delta\mathcal{E}_d\Delta$ for every outcome d .

The natural target-violation operators of the fermion controller are

$$\hat{D}_j = \frac{1 - \tau_j(2n_{\chi_j} - 1)}{2}. \quad (69)$$

They do not define simultaneous bits because

$$[n_{\chi_j}, n_{\chi_k}] = c^\dagger [P_j, P_k] c. \quad (70)$$

Two rank-one projectors commute only when their orbitals are orthogonal or collinear. The finite-support controller is built from overlapping, nonorthogonal modes, so successive resets are classical in different bases. Coherences in one basis feed populations in the next.

A two-mode counterexample makes the problem explicit. Let

$$\chi = \frac{a+b}{\sqrt{2}}, \quad \varphi = \frac{a-b}{\sqrt{2}}. \quad (71)$$

The one-particle states $|\chi\rangle\langle\chi|$ and $|\varphi\rangle\langle\varphi|$ have the same diagonal in the site basis. Resetting χ empty sends the first state to vacuum but leaves the second state occupied in φ . Therefore the site-basis diagonal does not determine the future diagonal. A delocalized exact band basis may diagonalize the target, but it forfeits the local cellular-automaton structure. For a nonzero Chern band, the absence of a translation-covariant exponentially localized orthonormal Wannier basis makes this locality–diagonality tension structural; it is not by itself a no-go theorem for every possible invariant diagonal algebra.

The conclusion is narrower than a no-go theorem for every conceivable classical representation. It rules out the direct O'Dea reduction to a local common-basis bitstring for the generic overlapping controller. The exact positive covariance-manifold transfer in Eq. (36) remains valid.

X. TOWARD AN EFFECTIVE WALL DEFECT THEORY

This section deliberately changes status from exact identities to candidate approximations. The purpose is to identify calculations that could turn the formal defect field into a useful low-energy theory.

A. Narrow-distribution closure

The simplest closure replaces the nonlinear noise source in Eq. (46) by its value on the deterministic mean orbit. Here and below,

$$\bar{g}_t = \chi_t^\dagger \bar{G}_{t-1} \chi_t, \quad \bar{R}_t = R_t(\bar{G}_{t-1}). \quad (72)$$

Thus the candidate closure is

$$\mathbb{E} \left[\frac{|R_t\rangle\langle R_t|}{1 - g_t^2} \right] \simeq \frac{|\bar{R}_t\rangle\langle \bar{R}_t|}{1 - \bar{g}_t^2}. \quad (73)$$

The result is a closed discrete Lyapunov equation for Σ_t . It is not controlled merely because the incoming state is Gaussian. It becomes plausible when the trajectory distribution is demonstrably narrow, or in a modified many-flavor model with a saddle suppressing collective fluctuations. The approximation can be tested directly by measuring both sides of Eq. (73).

B. Retarded click interaction from linearized covariance response

Assume $|\bar{g}_t| < 1$ and define $\bar{W}_t = \bar{R}_t/(1 - \bar{g}_t^2)$. We now make a distinct weak-noise closure: freeze the nonlinear

kick matrix $W_t(G) = R_t(G)/(1 - g_t^2)$ on the mean orbit and neglect its derivative. Retaining the first-order state dependence of the propensity gives

$$\mathcal{J}_t(H) = Q_t H Q_t + \bar{W}_t(\chi_t^\dagger H \chi_t), \quad \bar{q}_t = \frac{1 - \tau_t \bar{g}_t}{2}. \quad (74)$$

Indeed, $d_t - q_t = d_t - \bar{q}_t + (\tau_t/2)\chi_t^\dagger \delta G_{t-1} \chi_t$, so the fluctuation recursion in this closure is

$$\delta G_t \simeq \mathcal{J}_t(\delta G_{t-1}) + 2\tau_t(d_t - \bar{q}_t)\bar{W}_t. \quad (75)$$

For a single Bernoulli click, the discarded derivative of W_t is not parametrically small; Eq. (75) becomes controlled only if coarse graining, weak overlap, or a large-flavor construction supplies a small-noise parameter.

Let $\mathcal{J}_{t-1:u+1}$ denote the chronological product propagating a matrix from just after event u to just before event t , and define

$$\Gamma_{tu} = \chi_t^\dagger \mathcal{J}_{t-1:u+1}(\bar{W}_u) \chi_t \geq 0. \quad (76)$$

The inequality follows because $\mathcal{J}_t$ maps positive matrices to positive matrices. Within the stated closure, the first-order response is

$$q_t \simeq \bar{q}_t - \sum_{u < t} \tau_u \Gamma_{tu} (d_u - \bar{q}_u). \quad (77)$$

If $\bar{q}_t$ stays bounded away from 0 and 1, and the induced propensity shifts are small, expanding the Bernoulli log likelihood around $\bar{q}_t$ gives

$$\begin{aligned} \log p[\mathbf{d}] &\simeq \sum_t [d_t \log \bar{q}_t + (1 - d_t) \log(1 - \bar{q}_t)] \\ &- \sum_{t > u} \frac{\tau_u \Gamma_{tu}}{\bar{q}_t(1 - \bar{q}_t)} (d_t - \bar{q}_t)(d_u - \bar{q}_u) + \dots \end{aligned} \quad (78)$$

The complete-history quadratic form can be symmetrized algebraically over unordered spacetime pairs. Its “retarded” character refers instead to its causal construction, temporal anisotropy, and generic lack of time-reversal symmetry. Near $\bar{q}_t = 0$ or 1, a separate rare-click expansion is required. Nonlinear kick dependence and partner-orientation fluctuations generate higher-body terms.

Equation (78) suggests a concrete wall mechanism. If the homogeneous linearized response map contracts rapidly in the bulk, Γ_{tu} is short ranged there. A near-marginal wall mode makes the kernel long lived and possibly directional along the interface. A defect statistical mechanics could then emerge on the wall even though no local click-only rule exists microscopically. This is a conjecture until Γ_{tu} is measured and shown to predict click correlations.

C. Weak-overlap kinetics and deliberate twirls

There is also a geometric weak-overlap expansion. Suppose χ_j has just been reset to τ_j , and then the *unconditional* reset of χ_k to τ_k is applied. With $\alpha = \langle \chi_k | \chi_j \rangle$ and

$h = \chi_k^\dagger G \chi_k$, the mean re-created mismatch propensity at j is

$$q'_j = |\alpha|^2 \left[1 - \frac{\tau_j}{2}(h + \tau_k) \right]. \quad (79)$$

Thus nonorthogonality can re-create mismatch propensity at order $|\alpha|^2$, while the coefficient still remembers the hidden covariance. This identity motivates a kinetic approximation but also displays exactly why overlap weights alone do not close it.

An exact finite classical process can be manufactured by imposing a common dephasing channel, an appropriate Haar twirl, or a commuting/orthogonal controller limit. Those are modified models. They may destroy the coherent class-A wall and must not be presented as derivations of the original circuit. A large-flavor generalization is another possible route: replica interactions can admit a saddle and organize corrections in inverse flavor number, but independent uncoupled copies do not by themselves control the single-flavor problem.

XI. CONCRETE DIAGNOSTICS AND FALSIFICATION TESTS

The defect proposal becomes useful only if it produces tests beyond a relabeling of the measurement record.

1. *Exact small-system enumeration.* Enumerate every click string for small Fock-space systems. Compare $\mathcal{Z}_n$, the activity generating function, and integer-replica insertions with Eqs. (54) and (58).
2. *Controller conflict graph.* Measure $[P_j, P_k]$, cross-target overlaps, and the directed weights relevant to the chronological BPJ word. Test the floor in Eq. (27) event by event.
3. *Partner-mode propagation.* On pure trajectories, construct u_t and test Eq. (25) without fitted parameters. This directly asks whether click correlations follow the state-dependent partner overlap.
4. *Exact fluctuation source.* Measure the source term in Eq. (46) and compare it with the mean-orbit closure in Eq. (73). This quantifies rather than assumes moment closure.
5. *Retarded kernel.* Compute Γ_{tu} from the linearized response map and compare Eq. (77) with conditional click rates. Separate wall-on, wall-off, and homogeneous controls.
6. *Dynamical activity.* Use the exact tilted Gaussian recursion to obtain the activity SCGF, susceptibility, and signed-charge cumulants. Look for wall-excess finite-size structure only after subtracting matched bulk update counts.

7. *Replica fluctuation check.* Reconstruct the full second moment $M = \Sigma + |\langle G \rangle| \langle \bar{G} |$, and compare its contractions with direct trajectory estimates of $|\bar{C}_{ij}|^2$ and $\bar{F}_A^{(q)}$. Keep these distinct from quantities evaluated on $\bar{C}$.

Directed-percolation language is not justified by the existence of a binary click alone. It would require a genuine absorbing configuration, a closed short-range scalar defect kinetics, and the absence of a relevant conservation law or symmetry. The generic compact noncommuting controller may instead have intrinsic mismatch activity. Its signed activity obeys the exact charge accounting in Eq. (11) and has zero mean over a stationary cycle, but this is not itself a conservation law for the reduced system or the click kinetics. The appropriate starting hypothesis is therefore a driven shot-noise field coupled to slow wall covariance modes, not directed percolation by default.

XII. DISCUSSION

The mismatch-click variable does organize the physics, but in a more nuanced way than a classical occupation defect. At one event it is an exact binary innovation. In a pure trajectory it flips one state-dependent partner mode, giving a precise particle/hole interpretation and exact signed-charge accounting. Across events it is emitted by a hidden Markov process whose state is the covariance matrix. The corresponding positive transfer operator already supplies a legitimate classical statistical mechanics, while the replicated density transfer supplies a local fermionic tensor-network version.

The one-copy Born average is unusually simple because the nonlinear recoil cancels between the two branches. That cancellation does not survive generic replica tilts or same-record nonlinear observables. The exact fluctuation equation identifies the missing piece as rank-at-most-one covariance shot noise. This is why the averaged covariance can look smooth or gapped while squared trajectory correlators retain substantial wall structure.

The comparison with O’Dea *et al.* isolates the structural question. Their averaged dynamics preserves a common diagonal defect algebra, so a local classical cellular automaton is exact. Here the compact class-A controller uses overlapping noncommuting orbitals. A local common diagonal algebra is not available generically, and the correction event is immediately healed at its own orbital. The plausible emergent theory is consequently a wall-localized activity field with retarded interactions mediated by covariance modes, or an interacting replicated fermion theory with a defect insertion, rather than a microscopic reaction–diffusion process assumed from the outset.

The next decisive calculation is therefore not to fit an arbitrary classical model to click rasters. It is to evaluate the controller conflict graph, the exact partner-mode propagation weights, the nonlinear fluctuation source, and the

linearized response Green function on the same wall and wall-off geometries. Those objects determine whether the retarded action truncates to a useful local wall theory or whether the full replicated fermionic transfer must be retained.

Appendix A: One-event algebra

For a branch with nonzero Born probability, the Sherman–Morrison identity gives

$$(\mathbf{1} + \eta_m GP)^{-1} = \mathbf{1} - \frac{\eta_m GP}{1 + \eta_m g}. \quad (\text{A1})$$

Multiplication by Q on both sides yields

$$\Phi_{m|s}(G) = QGQ + \tau P - \frac{\eta_m}{1 + \eta_m g} R. \quad (\text{A2})$$

For $|g| < 1$, using $\eta_m = \tau(-1)^d$ and

$$-\frac{\eta_m}{1 + \eta_m g} = \frac{g - \eta_m}{1 - g^2} = \frac{2\tau[d - q_s(G)]}{1 - g^2} \quad (\text{A3})$$

proves Eq. (14).

The first conditional moment follows from $\mathbb{E}(d - q|G) = 0$. For the second,

$$\begin{aligned} \mathbb{E}_d[(\Phi_d - B) \otimes (\Phi_d - B)|G] &= \frac{4R \otimes R}{(1 - g^2)^2} \mathbb{E}[(d - q)^2|G] \\ &= \frac{R \otimes R}{1 - g^2}, \end{aligned} \quad (\text{A4})$$

which proves Eq. (44).

For a pure covariance, $G^2 = \mathbf{1}$ gives

$$\|QG\chi\|^2 = \chi^\dagger GQG\chi = 1 - g^2. \quad (\text{A5})$$

The relation $QGQ u = -gu$ follows by applying QG to $G\chi = g\chi + \sqrt{1 - g^2}u$. Substitution into Eq. (20) proves Eq. (21).

Appendix B: Tilted block identity and covariance

Write the density operator in blocks of the χ -mode occupation. The projective measurement removes off-diagonal blocks, and the conditional correction resets the exposed mode to s :

$$K_{d|s} \rho K_{d|s}^\dagger = |s\rangle\langle s|_\chi \widehat{\otimes}_\chi \langle s \oplus d | \rho | s \oplus d \rangle_\chi. \quad (\text{B1})$$

Multiplying by x^d and summing proves Eq. (50). The normalized covariance is the weighted branch average

$$\Psi_{s,x}(G) = \frac{p_0 \Phi_0 + x p_1 \Phi_1}{p_0 + x p_1}. \quad (\text{B2})$$

Since

$$p_0 \frac{-\tau}{1 + \tau g} = -\frac{\tau}{2}, \quad p_1 \frac{\tau}{1 - \tau g} = \frac{\tau}{2}, \quad (\text{B3})$$

Eq. (52) follows immediately.

Appendix C: Proof of the consecutive overlap bound

For two marks,

Suppose $G\chi = \tau\chi$ and decompose

$$\psi = \alpha\chi + w, \quad \langle \chi | w \rangle = 0, \quad \|w\|^2 = 1 - |\alpha|^2. \quad (\text{C1})$$

Hermiticity gives $\langle \chi | Gw \rangle = 0$, so

$$\psi^\dagger G\psi = \tau|\alpha|^2 + w^\dagger Gw. \quad (\text{C2})$$

For $\tau' = -\tau$ and $-1 \leq G \leq 1$,

$$\tau' \psi^\dagger G\psi \leq -|\alpha|^2 + (1 - |\alpha|^2) = 1 - 2|\alpha|^2. \quad (\text{C3})$$

Therefore

$$q_{\psi, \tau'}(G) = \frac{1 - \tau' \psi^\dagger G\psi}{2} \geq |\alpha|^2, \quad (\text{C4})$$

which proves Eq. (27).

Appendix D: Marked-replica bookkeeping

For a single marked observable,

$$\begin{aligned} \text{Tr}[O^{(1)}\Omega_n] &= \sum_{\xi} \text{Tr}(O\tilde{\rho}_{\xi})(\text{Tr}\tilde{\rho}_{\xi})^{n-1} \\ &= \sum_{\xi} p_{\xi}^n \langle O \rangle_{\xi}. \end{aligned} \quad (\text{D1})$$

Equations (63)–(65) follow by choosing the marked operators and continuing the normalized ratio to $n = 1$.

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